Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________

Factoring a Difference of Squares

Example (Factor the difference of squares):
$x^2 - 9$ — both are perfect squares (x^2 and 3^2) with a minus, so $x^2 - 9 = (x+3)(x-3)$.

Directions: Factor each. If it is not a difference of squares, write "prime."

1. $x^2 - 25$

2. $x^2 - 49$

3. $4x^2 - 1$

4. $9x^2 - 16$

5. $x^2 + 16$

Challenge Problems

6. $16x^2 - 81$ — factor completely.

Answer Key and Procedure

Procedure

1. Confirm both terms are perfect squares with a minus between them.
2. Take the square root of each term (a and b).
3. Write $(a + b)(a - b)$; a sum of squares is prime.

Answers

1. $(x + 5)(x - 5)$

2. $(x + 7)(x - 7)$

3. $(2x + 1)(2x - 1)$

4. $(3x + 4)(3x - 4)$

5. prime (sum of squares)

6. $(4x + 9)(4x - 9)$